

Monte Carlo Counterfactual Regret Minimization

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Abstract

This report studies Monte Carlo Counterfactual Regret Minimization (MCCFR), focusing on the external-sampling variant. The project builds on a previous implementation of full-tree CFR for Kuhn Poker and asks what changes when the regret update is estimated from sampled trajectories instead of a complete traversal of the game tree. I implement both full-tree CFR and external-sampling MCCFR in Python, compare them on Kuhn Poker, and use exact best responses to measure exploitability. The main message is that MCCFR keeps the same local regret-minimization structure as CFR, but replaces exact counterfactual values with sampled estimates.

1 From CFR to MCCFR

Counterfactual Regret Minimization (CFR) is a regret-minimization method for extensive-form games with imperfect information. The key idea is to place a small regret minimizer at every information set. At each iteration, the algorithm computes a counterfactual value for every legal action at each information set, updates local regrets, and then obtains the next behavioral strategy by regret matching.

In the full-tree version of CFR, the update is exact for the current strategy profile because it traverses the entire game tree. This is very clear pedagogically, but it is also the bottleneck: one iteration costs time proportional to the size of the tree. Kuhn Poker is tiny, so this is not an issue there. In larger games such as poker variants with more cards, betting actions, or public states, a full traversal can become too expensive.

MCCFR keeps the same local regret-minimization structure but estimates the counterfactual update by sampling part of the tree. The version studied here is *external sampling*. When updating player i , chance nodes and opponent nodes are sampled, while player i 's own decision nodes branch over all legal actions. Therefore, when the sampled path reaches an information set of player i , the algorithm can still compare all actions at that information set:

$$R_i^T(I, a) \leftarrow R_i^{T-1}(I, a) + \hat{v}_i(I, a) - \hat{v}_i(I),$$

where the hats indicate sampled estimates. The important point is that this has the same form as the CFR regret update, but the values are computed using a sampled subtree instead of the whole tree.

2 Kuhn Poker as the Test Game

Kuhn Poker is a small two-player zero-sum poker game introduced by Kuhn [3]. The deck contains three cards, usually denoted J, Q, K . Each player antes one chip and receives one private card. Player 0 acts first and can either check or bet. If player 0 checks, player 1 can check or bet. If a bet occurs, the other player can call or fold. If no one folds, the higher card wins the pot.

The game is small enough that every strategy and best response can be computed exactly, but it still contains the main modeling features that make poker interesting:

- private information, because each player only observes their own card;
- sequential decisions, because betting history affects later actions;
- information sets, because the same private card and public history can arise under different hidden opponent cards;
- bluffing, because weak hands sometimes bet in equilibrium.

The known value of Kuhn Poker for player 0 is

$$v^* = -\frac{1}{18} \approx -0.055556.$$

This value is useful as a sanity check: if an implementation converges to a strategy profile whose expected payoff is far from this value, then something is probably wrong.

In the code, the game is represented as a simple dictionary of nodes. A chance node stores the six equally likely deals:

$$KQ, QK, JK, KJ, QJ, JQ.$$

A decision node stores the acting player, the information set, and the legal actions. A terminal node stores the zero-sum utility vector. This is not intended to be a general game engine. The point is to make the algorithm readable and to keep the representation close to the game tree.

3 Regret Matching and Full-Tree CFR

Regret matching is the local learning rule used in this project. Suppose an information set has action set A . Let $R^T(a)$ be the cumulative regret for action a after T updates. Regret matching chooses the next mixed action by normalizing only the positive regrets:

$$\sigma^{T+1}(a) = \begin{cases} \frac{[R^T(a)]_+}{\sum_{b \in A} [R^T(b)]_+}, & \text{if } \sum_{b \in A} [R^T(b)]_+ > 0, \\ \frac{1}{|A|}, & \text{otherwise.} \end{cases}$$

The uniform fallback matters early in training because all regrets start at zero.

In full-tree CFR, each information set has its own regret table. At each iteration and for each update player i , the algorithm recursively traverses the full game tree. Let I be an information set of player i , and let $a \in A(I)$. The update has the form

$$R_i^T(I, a) \leftarrow R_i^{T-1}(I, a) + \pi_{-i}^{\sigma^T}(I) (v_i(\sigma_{I \rightarrow a}^T, I) - v_i(\sigma^T, I)).$$

Here $\pi_{-i}^{\sigma^T}(I)$ is the counterfactual reach probability from chance and the opponent, $v_i(\sigma_{I \rightarrow a}^T, I)$ is the counterfactual value of deviating to action a at I , and $v_i(\sigma^T, I)$ is the value of following the current mixed action at I .

For implementation, this can be written more concretely. At an update player node, recursively compute one child value for each legal action:

$$u(a) = \text{value returned by the child after action } a.$$

The value of the current mixed action is

$$u(\sigma) = \sum_{a \in A(I)} \sigma(I, a) u(a).$$

Then the regret increment is proportional to $u(a) - u(\sigma)$. The counterfactual reach factor is what makes the local regret update consistent with the global extensive-form game.

The full-tree implementation in this project keeps two tables: cumulative regrets and cumulative average-strategy weights. The regret table determines the current strategy through regret matching. The strategy-sum table is used at the end to return the average strategy, which is the strategy that converges in the standard CFR guarantee.

4 External Sampling

The implementation uses the following recursive view. At a terminal node, the recursion returns the payoff of the update player. At a chance node, one chance outcome is sampled. At an opponent node, one opponent action is sampled from the opponent's current regret-matching strategy. At an update player node, every legal action is recursively evaluated, and the local regret increment is

$$\hat{v}(a) - \sum_{b \in A(I)} \sigma(I, b) \hat{v}(b).$$

This is the same action-value minus node-value form used in ordinary CFR.

The average strategy is stored in a tabular strategy-sum table. For external sampling, I use the simple sampled averaging rule: when an opponent information set is visited during a sampled traversal, the opponent's current strategy at that information set is added to the corresponding strategy sum. Each player is the opponent once per MCCFR iteration because the implementation performs one update for player 0 and one update for player 1.

5 Implementation

The code is self-contained and intentionally small. The Kuhn Poker game is represented as a dictionary of extensive-form nodes. Chance nodes store the six equally likely card deals, decision nodes store player, information set, and action list, and terminal nodes store the zero-sum utility vector.

Both algorithms share the same basic interface:

```
train(iterations),    current_strategy(),    average_strategy().
```

The full-tree CFR class keeps the standard counterfactual reach-probability weights. The MCCFR class keeps the same regret tables but only traverses a sampled part of the tree during each update. I also implemented exact evaluation helpers for expected value, best response value, and exploitability. For exploitability, the behavioral strategy is converted to sequence-form realization weights and then evaluated against exact best responses.

The main source files are organized as follows:

- `kuhn_poker.py`: explicit Kuhn Poker game tree and a compact player-centric representation used for exact evaluation;
- `regret_matching.py`: positive-regret normalization with a uniform fallback;
- `cfr.py`: full-tree CFR baseline;
- `external_sampling_mccfr.py`: external-sampling MCCFR;
- `util.py`: expected value, best response, exploitability, and strategy formatting helpers.

Exploitability is computed exactly rather than estimated from rollouts. First, the behavioral strategy is converted into sequence-form realization weights. Then, for each player, the value of a best response to the opponent’s fixed strategy is computed by dynamic programming over that player’s tree-form sequential decision process. The reported exploitability is

$$\frac{1}{2} (\text{BR}_0(\bar{\sigma}_1) + \text{BR}_1(\bar{\sigma}_0)),$$

which is NashConv divided by two for a two-player zero-sum game.

I also count how many tree nodes are visited during regret updates. This counter is not needed by the algorithm, but it helps explain the difference between full-tree CFR and MCCFR. Full CFR visits the whole tree on each update. External sampling visits only a sampled part of the tree, although it still branches over the update player’s own actions.

6 Tests and Sanity Checks

The project includes small tests because implementation errors in CFR can be subtle. The tests check:

- the Kuhn Poker chance node has six uniform deals;
- all terminal utilities are zero-sum;
- regret matching normalizes positive regrets correctly and falls back to the uniform strategy when needed;
- exploitability is finite for a known simple strategy;
- full-tree CFR converges near the Kuhn Poker value after 10,000 iterations;
- external-sampling MCCFR reduces exploitability over multiple fixed random seeds;
- MCCFR visits fewer nodes per iteration than full-tree CFR.

These tests are not meant to prove correctness, but they catch the most common mistakes: wrong terminal payoffs, incorrect information-set action lists, missing uniform fallback, and regret updates with the wrong sign.

7 Experiments

The experiment compares full-tree CFR and external-sampling MCCFR on Kuhn Poker. The known value of Kuhn Poker for player 0 is

$$-\frac{1}{18} \approx -0.055556.$$

I ran full-tree CFR for 10,000 iterations and MCCFR for 100,000 iterations with three random seeds. The MCCFR numbers below are averaged over seeds.

Algorithm	Iterations	Value	Exploitability	Regret-update nodes
CFR	10,000	-0.055561	0.001982	1.10×10^6
ES-MCCFR	100,000	-0.055551	0.001746	1.26×10^6

Both methods converge close to the known value. CFR has a smoother curve because every update is exact for the current profile. MCCFR is noisier, but each iteration visits far fewer nodes. In this implementation, full CFR visits about 110 tree nodes per iteration, while external sampling visits about 12.6 nodes per iteration on average.

This node-count comparison is the main reason MCCFR is interesting. On Kuhn Poker, full CFR is already fast, so runtime alone is not a very meaningful benchmark. But the number of traversed tree nodes is a proxy for what would happen in a larger

extensive-form game: if one full iteration is too expensive, then it is useful to trade exact updates for many cheaper sampled updates.

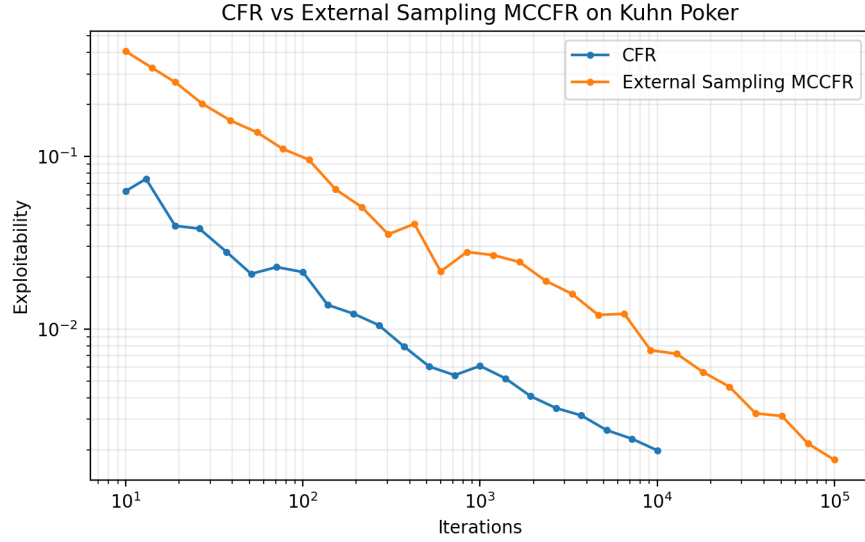


Figure 1: Exploitability of full-tree CFR and external-sampling MCCFR on Kuhn Poker.

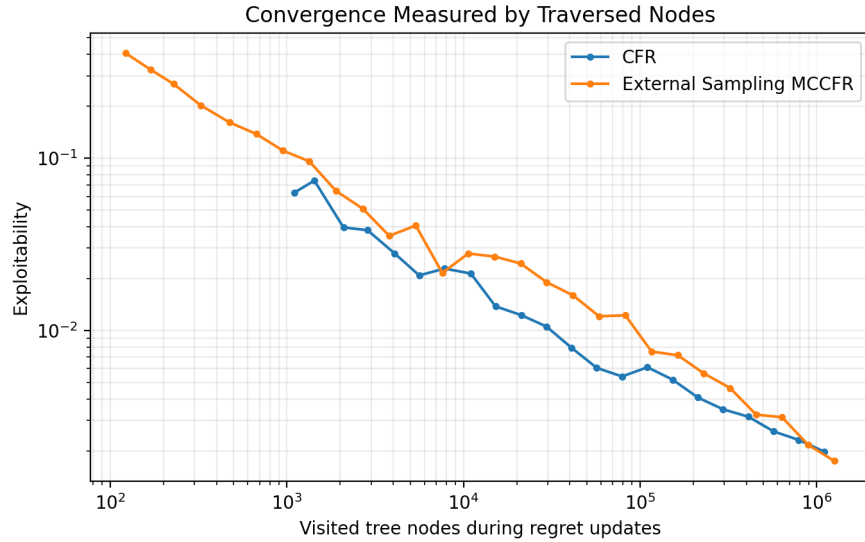


Figure 2: The same convergence curves measured by the number of tree nodes visited during regret updates.

The learned strategies also match the qualitative structure of Kuhn Poker equilibria.

Weak hands mostly check and fold against bets, strong hands bet or call aggressively, and the middle card mixes in response states. This is a useful sanity check because a small exploitability number alone does not always make implementation mistakes easy to spot.

8 Learned Strategies

The learned average strategies also give a qualitative sanity check. The following table shows selected information sets from one run. The exact numbers are not unique because Kuhn Poker has multiple equilibrium strategies, but the broad structure is correct: weak hands usually avoid calling, strong hands call or bet aggressively, and some bluffing appears with the weakest card.

Algorithm	Information set	First action prob.	Second action prob.
CFR	J (check, bet)	0.7727	0.2273
CFR	K (check, bet)	0.2978	0.7022
CFR	Q check-bet (call, fold)	0.5632	0.4368
MCCFR	J (check, bet)	0.7311	0.2689
MCCFR	K (check, bet)	0.2056	0.7944
MCCFR	Q check-bet (call, fold)	0.5979	0.4021

The most useful feature of this table is not the exact decimal values, but the pattern. With J , the weakest card, the strategy mixes in some bets as bluffs but usually checks and folds when facing bets. With K , the strongest card, the strategy bets frequently and calls bets almost always. The queen is the marginal card and therefore mixes in response states.

9 Discussion and Bibliography

The main lesson from this project is that MCCFR is not a different conceptual algorithm from CFR. It is better understood as the same local counterfactual regret update with a sampled estimator for the values. External sampling is especially easy to understand because the update player still branches over all actions at visited information sets.

The limitation of this project is that Kuhn Poker is small. Full-tree CFR is already cheap, so the practical scaling advantage of MCCFR is only visible through the node-visit counter rather than through a difficult large-game benchmark. Natural next steps would be outcome sampling, Leduc Poker, and abstraction methods such as bucketing.

External sampling is pedagogically attractive because it preserves the “compare all of my actions” step at each visited information set. This made the implementation close to the full-tree CFR baseline. The price is variance: a single sampled traversal sees only one chance outcome and one opponent action at each sampled opponent decision. Outcome sampling would go further by sampling actions for all players and using importance weighting. That can make each iteration even cheaper, but the estimator is usually higher variance and more delicate to implement correctly.

References

- [1] Martin Zinkevich, Michael Johanson, Michael Bowling, and Carmelo Piccione. Regret minimization in games with incomplete information. In *Advances in Neural Information Processing Systems 20*, 2007. <https://papers.nips.cc/paper/3306-regret-minimization-in-games-with-incomplete-information>.
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- [3] Harold W. Kuhn. A simplified two-person poker. In *Contributions to the Theory of Games*, 1950.